

Perfect competition, Cournot and Stackelberg

Consider a market with two firms, $i = 1, 2$, that compete à la Cournot. Each firm has profit function

$$\pi_i(q_i, q_{-i}) = [100 - q_i - q_{-i}] q_i - 40q_i$$

where q_i is firm i 's output and q_{-i} is the other firm's output.

1. Find the best-response functions, the Nash equilibrium quantities, total output, and each firm's profit. Suppose now that both firms can either comply with a collusive agreement, C , or not comply, NC . If both choose NC , the outcome is the Nash equilibrium from part (a). The payoff matrix is

	C	NC
C	(600, 450)	(200, 600)
NC	(800, 200)	—

2. Find the Nash equilibrium in pure strategies of this game. Identify whether the players have dominant strategies.
3. Graph the feasible payoff set and identify which payoffs can be sustained as average payoffs in the repeated game according to the Folk Theorem.
4. Find the minimum discount factor such that the grim-trigger strategy leads both firms to play C in every period of the infinitely repeated game. Keep in mind that this game is not symmetric.

Solution

1. The profit function of firm i is

$$\pi_i(q_i, q_{-i}) = (100 - q_i - q_{-i})q_i - 40q_i = (60 - q_i - q_{-i})q_i$$

For firm 1,

$$\pi_1(q_1, q_2) = 60q_1 - q_1^2 - q_1q_2$$

The first-order condition is

$$\frac{\partial \pi_1}{\partial q_1} = 60 - 2q_1 - q_2 = 0$$

Therefore,

$$BR_1(q_2) = \frac{60 - q_2}{2}$$

For firm 2,

$$\pi_2(q_1, q_2) = 60q_2 - q_2^2 - q_1q_2$$

The first-order condition is

$$\frac{\partial \pi_2}{\partial q_2} = 60 - q_1 - 2q_2 = 0$$

Therefore,

$$BR_2(q_1) = \frac{60 - q_1}{2}$$

The Cournot-Nash equilibrium satisfies

$$q_1 = \frac{60 - q_2}{2} \quad q_2 = \frac{60 - q_1}{2}$$

By symmetry,

$$q_1 = q_2 = q$$

Then,

$$q = \frac{60 - q}{2} \iff 2q = 60 - q \iff 3q = 60 \iff q = 20$$

Thus,

$$q_1^* = 20 \quad q_2^* = 20 \quad Q^* = 40$$

The market price is

$$P^* = 100 - Q^* = 100 - 40 = 60$$

Each firm's profit is

$$\pi_i^* = (P^* - 40)q_i^* = (60 - 40)20 = 400$$

Therefore, the missing payoff in the matrix is

$$(NC, NC) = (400, 400)$$

Conclusion: the Cournot-Nash equilibrium is

$$q_1^* = 20 \quad q_2^* = 20 \quad Q^* = 40 \quad P^* = 60$$

and each firm obtains

$$\pi_1^* = \pi_2^* = 400$$

2. Using the result from part (a), the one-shot collusion game is

	C	NC
C	$(600, 450)$	$(200, 600)$
NC	$(800, 200)$	$(400, 400)$

For firm 1, if firm 2 plays C , then

$$\pi_1(C, C) = 600 \quad \pi_1(NC, C) = 800$$

So firm 1 prefers NC . If firm 2 plays NC , then

$$\pi_1(C, NC) = 200 \quad \pi_1(NC, NC) = 400$$

So firm 1 also prefers NC . Hence, NC strictly dominates C for firm 1.

For firm 2, if firm 1 plays C , then

$$\pi_2(C, C) = 450 \quad \pi_2(C, NC) = 600$$

So firm 2 prefers NC . If firm 1 plays NC , then

$$\pi_2(NC, C) = 200 \quad \pi_2(NC, NC) = 400$$

So firm 2 also prefers NC . Hence, NC strictly dominates C for firm 2.

Conclusion: both firms have NC as a strictly dominant strategy. Therefore, the unique Nash equilibrium in pure strategies is

$$(NC, NC)$$

with payoffs

$$(400, 400)$$

3. The feasible payoff set is the convex hull, that is, the convex envelope, of the payoff vectors generated by the reduced 2×2 stage game:

$$(600, 450) \quad (200, 600) \quad (800, 200) \quad (400, 400)$$

The minmax payoff of firm 1 is 400. This is the lowest payoff firm 2 can force on firm 1, assuming that firm 1 best responds. To see this, let firm 2 play C with probability y . Firm 1's expected payoffs are

$$u_1(C) = 600y + 200(1 - y) = 200 + 400y$$

$$u_1(NC) = 800y + 400(1 - y) = 400 + 400y$$

Since NC gives a higher payoff for every y , firm 1's best response is NC . Firm 2 minimizes firm 1's best-response payoff by choosing $y = 0$, giving

$$v_1 = 400$$

Similarly, let firm 1 play C with probability x . Firm 2's expected payoffs are

$$u_2(C) = 450x + 200(1 - x) = 200 + 250x$$

$$u_2(NC) = 600x + 400(1 - x) = 400 + 200x$$

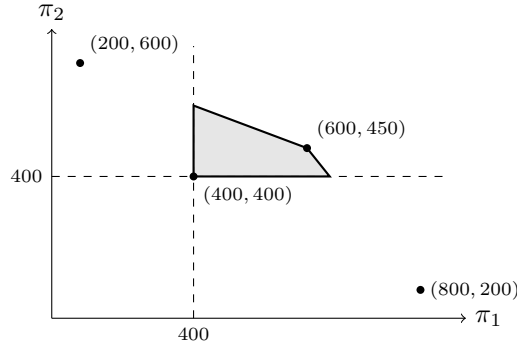
Since NC gives a higher payoff for every x , firm 2's best response is NC . Firm 1 minimizes firm 2's best-response payoff by choosing $x = 0$, giving

$$v_2 = 400$$

Therefore, the individually rational payoffs must give each firm at least its minmax payoff:

$$\pi_1 \geq 400 \quad \pi_2 \geq 400$$

The feasible and individually rational region is the part of the convex hull that lies weakly above both minmax payoffs.



By the Folk Theorem, in the infinitely repeated game, every feasible payoff vector that gives each player strictly more than his minmax payoff can be sustained as an average payoff for a sufficiently high discount factor. Therefore, the sustainable payoffs satisfy

$$(\pi_1, \pi_2) \in \text{co}\{(600, 450), (200, 600), (800, 200), (400, 400)\}$$

and

$$\pi_1 > 400 \quad \pi_2 > 400$$

Conclusion: the payoffs that can be sustained by the Folk Theorem are the feasible payoff vectors that are strictly individually rational:

$$\pi_1 > 400 \quad \pi_2 > 400$$

inside the convex hull of the stage-game payoff vectors

4. Under grim-trigger strategies, both firms cooperate as long as nobody has deviated in the past. If any firm deviates, both firms play the one-shot Nash equilibrium (NC, NC) forever after.

For firm 1, cooperating forever gives

$$\frac{600}{1 - \delta}$$

If firm 1 deviates while firm 2 cooperates, firm 1 obtains 800 today and then receives the punishment payoff 400 forever:

$$800 + \frac{\delta 400}{1 - \delta}$$

Thus, firm 1's incentive constraint is

$$\frac{600}{1 - \delta} \geq 800 + \frac{\delta 400}{1 - \delta}$$

Multiplying by $1 - \delta$,

$$600 \geq 800(1 - \delta) + 400\delta$$

$$600 \geq 800 - 400\delta \iff 400\delta \geq 200 \iff \delta \geq \frac{1}{2} = 0.5$$

For firm 2, cooperating forever gives

$$\frac{450}{1-\delta}$$

If firm 2 deviates while firm 1 cooperates, firm 2 obtains 600 today and then receives the punishment payoff 400 forever:

$$600 + \frac{\delta 400}{1-\delta}$$

Thus, firm 2's incentive constraint is

$$\frac{450}{1-\delta} \geq 600 + \frac{\delta 400}{1-\delta}$$

Multiplying by $1-\delta$,

$$450 \geq 600(1-\delta) + 400\delta$$

$$450 \geq 600 - 200\delta \iff 200\delta \geq 150 \iff \delta \geq \frac{3}{4} = 0.75$$

Since both constraints must hold, the minimum discount factor is

$$\delta \geq \max \left\{ \frac{1}{2}, \frac{3}{4} \right\} = \frac{3}{4} = 0.75$$

Conclusion: grim-trigger strategies sustain cooperation in every period if and only if

$$\delta \geq \frac{3}{4} = 0.75$$